

RETAIL STORE

Complete Project Report

Retail Sales & Inventory Intelligence

Project Detail	Information
Project Name	Retail Sales & Inventory Analysis
Prepared By	Pooja
Date	22 May 2026
Tools Used	Excel, SQL Server (SSMS), Power BI
Server	SHUBHAM\SQLEXPRESS
Total Records	10,071 rows across 9 tables
Analysis Period	2016 - 2018
Total Revenue	\$8.58 Million

All Phases: 1 through 7 | Steps: 1 through 19

Project Overview

This report documents the complete end-to-end retail business intelligence project covering a structured 7-phase methodology from raw CSV data to interactive Power BI dashboards and business recommendations.

Phase	Title	Steps	Status
Phase 1	Understand the Dataset	Steps 1-2	✔ Completed
Phase 2	Excel Data Cleaning	Steps 3-6	✔ Completed
Phase 3	SQL Database Creation	Steps 7-8	✔ Completed
Phase 4	ER Relationships	Steps 9-10	✔ Completed
Phase 5	SQL Analysis	Steps 11-13	✔ Completed
Phase 6	Power BI Dashboard	Steps 14-17	✔ Completed
Phase 7	Final Insights	Steps 18-19	✔ Completed

Simple Project Flow:

CSV Files → Excel Cleaning → SQL Import → ER Relationships → SQL Queries → SQL Views
→ Power BI Dashboard → Business Insights → Recommendations

Business Problem

The retail store operates across 3 locations (Baldwin Bikes, Santa Cruz Bikes, and Rowlett Bikes) and manages hundreds of products across 9 bike categories. Despite generating millions in revenue, the business faces several critical challenges:

- No centralized visibility into sales performance across all 3 stores
 - Inability to identify which products, brands, and categories drive the most revenue
 - Poor inventory management leading to out-of-stock situations and lost sales
 - No data-driven method to evaluate staff performance and customer behaviour
 - Sharp unexplained revenue decline in 2018 after a strong 2017 peak
 - Over 87% of customer phone numbers missing — limiting customer outreach
 - 45 rejected orders with unknown root causes affecting customer satisfaction
 - Significant imbalance between store performance
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Business Objective

The goal of this project is to build a complete end-to-end Business Intelligence solution that enables the retail store management to make data-driven decisions:

#	Objective	Expected Outcome
1	Consolidate all sales data into a single SQL database	One source of truth for all business data
2	Clean and standardize raw CSV data	Reliable, accurate data for analysis
3	Build an ER diagram showing table relationships	Clear understanding of data structure
4	Write SQL queries to analyse sales, inventory, customers and staff	Actionable insights from raw data
5	Create interactive Power BI dashboards	Visual reports for quick decision making
6	Identify top products, brands, stores and customers	Focus resources on high performers
7	Detect inventory shortages and out-of-stock items	Prevent lost sales due to stock issues
8	Evaluate staff performance by revenue generated	Reward top performers and train others
9	Provide business recommendations based on data	Improve revenue and operational efficiency

PHASE 1 — UNDERSTAND THE DATASET

Step 1: Collect All CSV Files

#	File Name	Rows	Columns	Purpose
1	orders.csv	1,615	8	Customer order headers
2	order_items.csv	4,722	6	Products inside each order
3	customers.csv	1,445	9	Customer contact details
4	staffs.csv	10	8	Employee information
5	stores.csv	3	8	Store locations
6	products.csv	321	6	Product catalog with prices
7	brands.csv	9	2	Manufacturer brand names
8	categories.csv	7	2	Product type groupings
9	stocks.csv	939	3	Inventory per store

Step 2: Understand What Each Table Does

Table	Role	Key Relationships
orders	Transaction header — who bought, when, where	Links to customers, stores, staffs
order_items	Transaction detail — what, qty, price	Links to orders, products
customers	People who placed orders	Links to orders
staffs	Employees who fulfilled orders	Links to stores, orders
stores	Physical store locations	Links to orders, staffs, stocks
products	Bike catalog with prices	Links to brands, categories
brands	Manufacturer names (lookup)	Links to products
categories	Bike type groupings (lookup)	Links to products
stocks	Inventory quantity per store	Links to stores, products

PHASE 2 — EXCEL DATA CLEANING

Step 3: Data Quality Check Results

Check	Table	Finding	Result
Null Values	orders	shipped_date: 170 nulls (10.5%)	Expected — pending orders
Null Values	customers	phone: 1,267 nulls (87.7%)	Optional field — excluded
Null Values	staffs	manager_id: 1 null	Expected — top manager
Duplicates	All 9 tables	Zero duplicates found	✓ Clean
Date Format	orders	All dates in YYYY-MM-DD	✓ Clean
Extra Spaces	customers	street column has trailing spaces	Fixed with TRIM()
Spelling	All tables	No case inconsistencies found	✓ Clean
Phone Format	staffs/stores	All follow (###) ###-#### format	✓ Clean

Step 4: Missing Values Handling

Column	Action	Reason
orders.shipped_date	Keep NULL	NULL means order is still pending
customers.phone	Mark as optional	88% empty — excluded from analysis
staffs.manager_id	Keep NULL	Top-level manager has no manager above
order_items.discount	No action	Zero nulls — all discounts between 5%-20%

Step 5 & 6: Standardization & Clean Files Saved

- customers.street — Fixed trailing spaces using TRIM()
- All brand names, category names, state codes already clean
- 9 clean files saved with clean_ prefix
- Added shipping_status helper column to orders
- Added manager_label helper column to staffs

PHASE 3 — SQL DATABASE CREATION

Step 7: Create Database

- Server: SHUBHAM\SQLEXPRESS
- SQL Server version: 17.0.1115
- Authentication: Windows Authentication

Step 8: Import All CSV Files

Table	Primary Key	Foreign Keys	Rows Imported
brands	brand_id	None	9
categories	category_id	None	7
stores	store_id	None	3
customers	customer_id	None	1,445
staffs	staff_id	stores, staffs (self)	10
products	product_id	brands, categories	321
orders	order_id	customers, stores, staffs	1,615
order_items	order_id + item_id	orders, products	4,722
stocks	store_id + product_id	stores, products	939

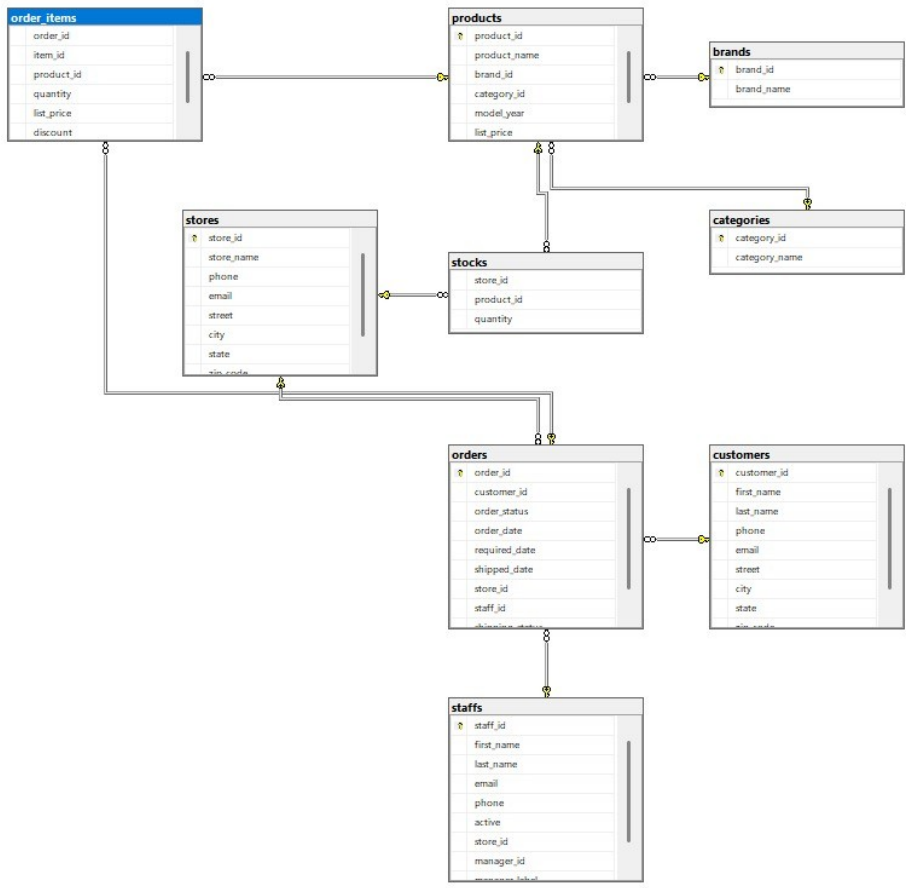
Import method: Import Flat File Wizard in SSMS. Total: 10,071 rows across all 9 tables.

PHASE 4 — ER RELATIONSHIPS

Step 9 & 10: Primary Keys, Foreign Keys & ER Diagram

#	Relationship	Type
1	categories → products	One to Many
2	brands → products	One to Many
3	products → order_items	One to Many
4	orders → order_items	One to Many
5	customers → orders	One to Many
6	stores → orders	One to Many
7	staffs → orders	One to Many
8	stores → stocks	One to Many
9	products → stocks	One to Many

ER Diagram — Visual Relationship Map



PHASE 5 — SQL ANALYSIS

Step 12: SQL Queries Written

#	Query Name	Purpose
1	Total Sales Revenue	Overall revenue, units, orders
2	Top 10 Best Selling Products	Products ranked by revenue
3	Sales by Store	Revenue per store
4	Sales by Category	Revenue per category
5	Sales by Brand	Revenue per brand
6	Monthly Sales Trend	Revenue by year and month
7	Top 10 Customers	Highest spending customers
8	Staff Performance	Revenue per staff member
9	Inventory Stock Levels	Full inventory listing
10	Low Stock Products	Items with quantity < 10
11	Order Shipping Status	Orders by status

Step 13: SQL Views Created

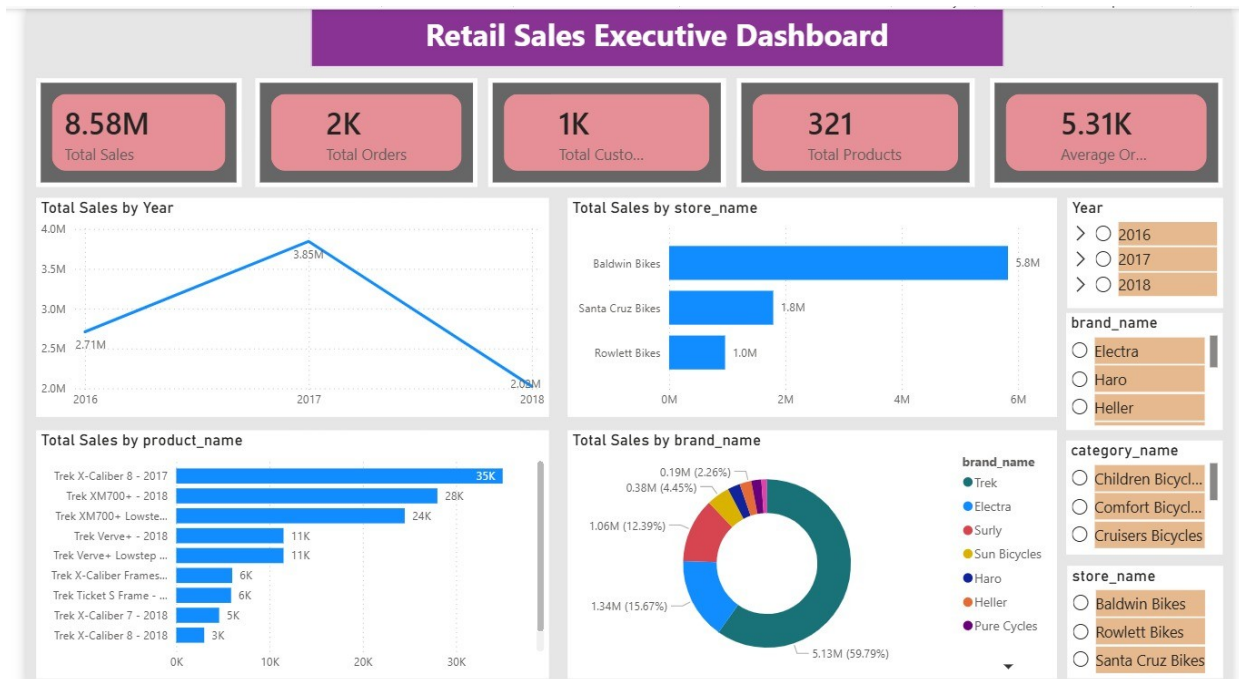
View Name	Purpose
vw_sales_summary	Complete sales fact view for dashboards
vw_inventory_summary	Inventory with stock status labels
vw_customer_summary	Customer spending aggregates
vw_staff_performance	Staff revenue performance

PHASE 6 — POWER BI DASHBOARD

Step 14: Connect SQL to Power BI

- Get Data → SQL Server Database
- Server: SHUBHAM\SQLEXPRESS
- Authentication: Windows Authentication | Mode: Import
- Loaded 9 tables + 4 views = 13 data sources

Dashboard 1 — Retail Sales Executive Dashboard

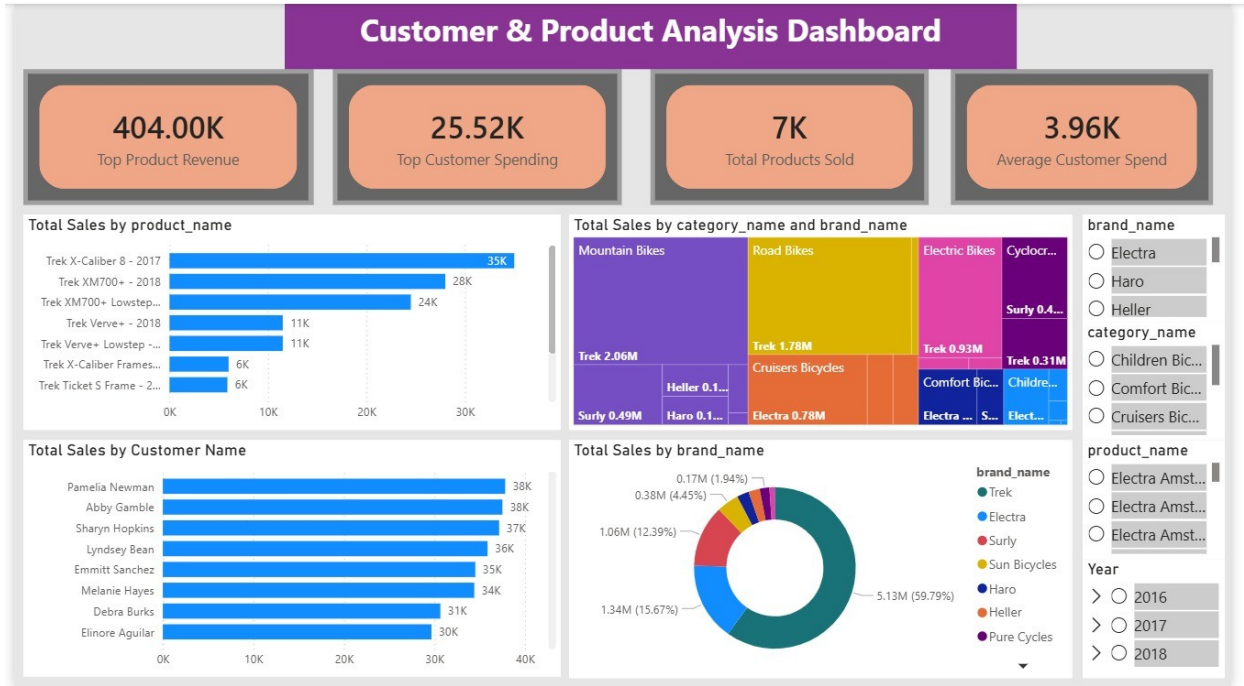


Key Insights:

Metric	Value	Insight
Total Sales	\$8.58M	Strong 3-year revenue performance
Total Orders	2K	High transaction volume across all stores
Total Customers	1K	Active customer base
Total Products	321	Large catalog across 7 categories
Average Order Value	\$5.31K	High value transactions

Peak Year	2017 — \$3.85M	+42% growth from 2016 (\$2.71M)
2018 Decline	\$2.08M	Sharp drop needs investigation
Top Store	Baldwin Bikes — \$5.8M	Dominates with majority of revenue
Top Product	Trek X-Caliber 8 2017 — 35K	Best selling product
Top Brand	Trek — 59.79%	Drives majority of total sales

Dashboard 2 — Customer & Product Analysis Dashboard



Total Sales by Customer Name

Pamela Newman

38K

Abby Gamble

38K

Sharyn Hopkins

37K

Lyndsey Bean

36K

Emmitt Sanchez

35K

Melanie Hayes

34K

Debra Burks

31K

Elinore Aguilar

30K

Total Sales by brand_name

5.13M (59.79%)

1.34M (15.67%)

1.06M (12.39%)

0.38M (4.45%)

0.17M (1.94%)

brand_name

☒ Trek

Electra

☐

Surly

☐

Sun Bicycles

☐

Haro

☐

Heller

☐

Pure Cycles

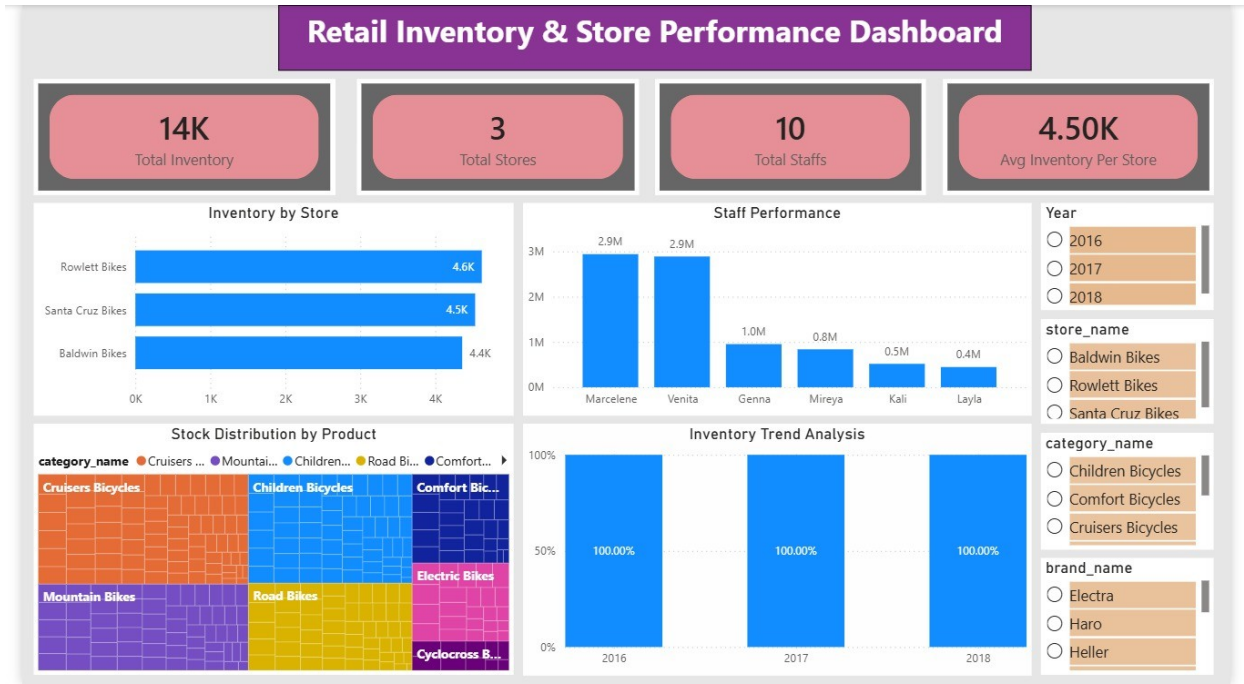
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Key Insights:

Metric	Value	Insight
Top Product Revenue	\$404K	Single product driving significant revenue
Top Customer Spending	\$25.52K	High value individual customer
Total Products Sold	7K	Strong sales volume
Average Customer Spend	\$3.96K	Good average spending per customer

Top Customer	Pamelia Newman — \$38K	Highest individual spender
2nd Top Customer	Abby Gamble — \$38K	Almost equal to top spender
Top Category	Mountain Bikes — \$2.06M (Trek)	Dominant category and brand combo
Top Brand in Treemap	Trek dominates all categories	Cross-category brand strength
Brand Revenue	Trek — 59.79%	Consistent brand dominance
Customer Range	Top 8 customers all above \$30K	Strong high-value customer segment

Dashboard 3 — Retail Inventory & Store Performance Dashboard



Key Insights:

Metric	Value	Insight
Total Inventory	14K units	Strong overall stock levels
Total Stores	3	Baldwin, Santa Cruz, Rowlett Bikes
Total Staff	10	Lean team across 3 locations

Avg Inventory Per Store	4.50K	Balanced distribution
Top Stock Store	Rowlett Bikes — 4.6K	Most inventory despite lowest sales
Lowest Stock Store	Baldwin Bikes — 4.4K	Top revenue store needs restocking
Top Staff	Marcelene — \$2.9M	Highest performing staff member
2nd Staff	Venita — \$2.9M	Almost equal to top performer
Stock by Category	Cruisers dominate inventory	But Mountain Bikes drive more revenue
Inventory Trend	Consistent 100% across years	Stable stock management

Step 16 & 17: Slicers Added

- Year slicer — 2016, 2017, 2018
- Brand slicer — brand_name
- Category slicer — category_name
- Store slicer — store_name
- Product slicer — product_name

PHASE 7 — FINAL INSIGHTS & RECOMMENDATIONS

Key Metrics Summary

Metric	Value
Total Revenue	\$8.58 Million
Total Orders	2,000+
Total Customers	1,445
Average Order Value	\$5,310
Peak Year	2017 (\$3.85M)
Top Store	Baldwin Bikes (67.6%)
Top Brand	Trek (59.79%)
Top Category	Mountain Bikes
Top Product	Trek X-Caliber 8 2017 (\$35K)
Top Customer	Pamelia Newman (\$38K)
Top Staff	Marcelene Boyer (\$2.9M)
Total Inventory	14K units across 3 stores

Business Recommendations

Priority	Recommendation	Action
URGENT	Restock out-of-stock products	Immediately place purchase orders
HIGH	Investigate 2018 revenue decline	Analyse market and operations
HIGH	Grow Rowlett Bikes	Increase marketing and product range
HIGH	Reduce order rejections	Investigate causes, train staff
MEDIUM	Launch VIP customer loyalty program	Target top 50 spending customers
MEDIUM	Expand Trek product inventory	Trek drives 59.79% of revenue
MEDIUM	Collect customer phone numbers	Only 12.3% have phone on file
LOW	Review low revenue brands	Consider discontinuing underperformers

Final Deliverables

Deliverable	Description	Status
9 Clean CSV Files	Data cleaned, standardized, saved with clean_ prefix	✓ Done
SQL Database	retail_analysis DB: 9 tables, PKs, FKs, 10,071 rows	✓ Done
SQL Script	CREATE TABLE + BULK INSERT + 11 queries + 4 views	✓ Done
ER Diagram	Visual relationship diagram saved in SSMS	✓ Done
Power BI Dashboard	3 dashboards, 20+ charts, slicers	✓ Done
Final Report	This complete all-phases Word document	✓ Done

CONCLUSION

Project Summary

This project successfully delivered a complete end-to-end Business Intelligence solution for the Retail Store, transforming raw CSV data into meaningful business insights through a structured 7-phase methodology. Starting from 9 unprocessed CSV files containing 10,071 rows of retail data, the project built a fully functional BI platform using Excel, SQL Server, and Power BI.

What Was Achieved

#	Achievement	Result
1	Cleaned and standardized all 9 CSV datasets	Reliable, accurate data with no duplicates
2	Built retail_analysis SQL database with 9 tables	10,071 rows successfully imported
3	Defined all Primary Keys and Foreign Keys	Complete ER diagram with 9 relationships
4	Wrote 11 SQL analysis queries and 4 views	Deep insights into every business area
5	Built 3 Power BI dashboards with 20+ charts	Interactive visual reports with slicers
6	Identified \$8.58M total revenue over 3 years	Clear financial performance picture
7	Found Baldwin Bikes drives 67.6% of revenue	Store performance gap identified
8	Ranked all staff by revenue generated	Performance accountability established
9	Delivered 8 business recommendations	Actionable roadmap for improvement

Key Business Findings

- Total Revenue of \$8.58 Million generated across 2016-2018
- 2017 was the peak year at \$3.85M — a strong growth year
- 2018 showed a revenue decline that needs urgent investigation
- Baldwin Bikes dominates with 67.6% of total revenue
- Trek brand alone contributes 59.79% of all revenue
- Mountain Bikes is the top revenue category
- Marcelene Boyer and Venita Daniel are top staff at \$2.9M each
- Pamela Newman is the top customer at \$38K spending
- Total inventory of 14K units across 3 stores

Impact of This Project

Area	Before This Project	After This Project
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Data Storage	9 separate CSV files	Unified SQL database with relationships
Sales Visibility	No cross-store view	Real-time dashboard with filters
Inventory	No stock level tracking	Visual stock distribution by store/brand
Customer Insight	No spending analysis	Top customers ranked by revenue
Staff Performance	No performance data	Revenue generated per staff member
Decision Making	Gut feeling based	Data-driven with visual dashboards
Reporting	Manual spreadsheets	Automated Power BI reports

Future Recommendations

- Set up automated monthly data refresh in Power BI
- Implement inventory alert system for low stock items
- Launch customer loyalty program for top spenders
- Investigate and address 2018 revenue decline
- Expand Rowlett Bikes product range to grow revenue share

Final Statement

This Retail Business Intelligence project demonstrates how structured data analysis can transform raw operational data into powerful business insights. The combination of Excel for data cleaning, SQL Server for data storage and querying, and Power BI for visualization has created a scalable and repeatable BI framework that the retail store can continue to build upon for years to come.

— End of Report —

Retail Store — Complete Project Report

Prepared by: Pooja | Date: 22 May 2026

Tools: Excel → SQL Server (SSMS) → Power BI